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A Self-Supervised Framework for Annotation-Free Lung Nodule Detection and Classification in CT Scans

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ABSTRACT
Lung cancer is one of the leading causes of cancer-related deaths worldwide, mainly due to its high incidence and low survival rates caused by late diagnosis. Thus, early detection of nodules is crucial for improving patient outcomes. Recent advancements in machine learning and deep learning have significantly boosted the accuracy of detecting lung nodules in CT scans. Given the high cost and scarcity of annotated CT data, creating an efficient, label-free computer-aided diagnosis (CAD) system is essential and particularly challenging. To tackle this, we present a self-supervised framework for detecting lung nodules without labeled data. This strong architecture derives hand-made descriptors (GLCM-based Haralick and Hu moments) and deep feature maps using a ResNet-50 backbone of all CT slices. These features are used to compute similarity scores at the pairwise level, and pseudo-labels of potential nodule regions are generated in a self-supervised pre-training phase. During the fine-tuning step, we optimize our ROI detector with the use of normalized cross-correlation (NCC) to relate fused feature templates with incoming slices, to create a similarity-based score map that indicates potential nodules. A dynamically adaptive step of K-means clustering sorts coherent NCC peaks into object propositions and optimizes each cluster by template matching as a refinement to increase localization and detection accuracy. In the end, a combination of handcrafted and deep representations of these proposals is ingested into any number of classifiers to produce an annotation-free pipeline that does not need manual labels to be trained. At the same time, limited ground-truth annotations were only used for template matching. Experimental results demonstrated the model's effectiveness in detecting and classifying lung nodules using self-supervised learning from CT scans, achieving 95.43% detection accuracy on the LUNA16 dataset and significantly reducing false positives. Findings illustrate how our model performs compared to current techniques.
Keywords: Lung Nodule Detection; Normalized Cross-Correlation; K-means Clustering; Self-Supervised Learning, Machine Learning; Computer Vision; Medical Image Analysis

1. INTRODUCTION

Lung cancer has been one of the most severe health issues in the world with the related deaths counting about 1.8 million each year with nearly 18 percent of all cancer-related deaths in the world [1]. According to the Global Cancer Observatory (GLOBOCAN), globally, lung cancer is the cause of the highest rates of deaths [2]. The incidence rate in Asia is 63.1 with a mortality rate of 62.9. It is estimated that the new cancer diagnoses in the world will amount to 2.5 million in 2024 which will comprise of 12.4 percent of the total cases [2]. Nonetheless, screening is not widely accepted, with only 16.0 percent of the eligible persons undergoing screening in 2022 [3]. The early opportunity to notice nodules in the lungs is a key to improved patient outcomes. Computer-aided diagnosis (CAD) systems are capable of helping radiologists, and computed tomography (CT) remains the gold standard in the identification and categorization of lung nodules [4-7]. Nevertheless, traditional image-processing methods are not effective in complex anatomical structures [8], and deep learning-based CAD systems cannot operate reliably without huge, annotated datasets [9]. This manually-based labeling creates a great bottleneck since the process of annotating 3D medical images is both expensive and time-consuming, and can be affected by inter-observer variability. This has resulted in an urgent necessity for label-free methods that decrease the reliance on labels and preserve the accuracy of diagnosis. Recent developments in self-supervised learning (SSL) have created potential pathways, and self-supervised models can learn high-resolution feature representations on unlabeled data, which in turn offers a scalable strategy to detect lung nodules [10-11].

A number of deep learning models have been suggested to classify lung nodules, including attention-based network models, such as the Progressive Growing Channel Attentive Non-Local (ProCAN) [12] network, specifically designed to classify nodules. So, by including a channel-wise attention mechanism [13] to the ResNet50 architecture by He et al. [14], the features are better represented, and the model is trained to improve the ability to localize nodules on CT scans using self-supervised learning methods. These have been preprocessed in terms of image normalization and ROI segmentation [15]. Dosovitskiy et al. suggested an ensemble method, which is a combination of GoogleNet, AlexNet, LeNet, ResNet, DenseNet, and VGG16, and an attention-based model [16]. Though better than other methods in feature representation and classification accuracy, the main commonality between them is that (1) they are very expensive and difficult to obtain large-scale, labeled datasets in medical imaging, and (2) they have a high computational cost when using high-resolution volumetric CT data. This difficulty is the reason why alternative approaches that are less reliant on annotation and are more computationally efficient are also of interest, prompting us to consider self-supervised learning as a method of lung nodule detection.

Self-supervised learning (SSL) [17] has emerged as an effective approach to pretraining on large unsupervised datasets that permits models to learn high-quality, domain-independent feature representations, which transfer to novel domains. Contrastive objective-based approaches using masked image pretraining have demonstrated considerable progress in medical image analysis, notably on pulmonary nodule detection using the LUNA16 challenge dataset [18]. Nevertheless, these approaches are data-heavy, which poses a training and generalization challenge. The cost of creating detailed labels to train deep learning models in 3D medical images is time-consuming and costly. The unlabeled datasets, which are used to minimize the ground truth labels that are elaborated, have been suggested by the use of the SSL methods [19]. To date, the majority of current 3D medical imaging techniques have performed simple pretext tasks on unlabeled data, on which the training of SSL methods is performed. In addition, the majority of the current research in the field of SSL is dedicated to general segmentation or domain adaptation, and less attention is paid to lesion-level identification and nodule characterization, which are of clinical importance in the early diagnosis of lung cancer. This weakness shows the necessity of an annotation-free SSL system that can not only help to decrease the use of ground-truth labels but also offer precise localization and classification of lung nodules.

Although development of CAD systems has been made, medical image processing in detecting nodules in the lungs is still difficult, as the sensitivity is low and false rates are high. We would like to propose an annotation-free self-supervised framework to overcome these problems and rely considerably less on labeled data, but keep the same level of detection accuracy. We combine descriptors based on handcrafted (GLCM-based Haralick and Hu moments) and deep representations of a ResNet-50 backbone to enhance the localization and classification of nodules on the LUNA16 dataset. The main works of this work can be summed up as:

1. To design a complete pipeline for detecting and classifying lung nodules using the Self-Supervised learning method.
2. To boost the performance in the detection and classification of lung CT images with normalized cross-correlation-based k-means clustering.
3. A combination of features-based method with traditional features (Haralick texture and Hu-Moment shape) and deep features showed better accuracy, and at the same time, low computational cost.

2. LITERATURE REVIEW

Achievement in identifying and classifying lung nodules. Resentment in detecting and classifying lung nodules using volumetric convolutional neural networks (3D CNNs), transformer-based self-supervised learning (SSL) methods, and LUNA16-specific SSL methodology. To provide an example, 3D U-Net and V-Net architecture were used by Ilesanmi et al. to learn how to capture relationships between slices in CT volumes and improve the accuracy of the segmentation process based on the fact that the segments should be spatially continuous [20]. On the same note, the DHEA-Net was built by Usman et al. as a dual-encoder model that fuses both side and front-view to achieve improved outcomes [21]. The common limitations of these volumetric CNN models are that, although they work, they require large, annotated datasets, as well as large amounts of GPU memory, which limits their applicability in data-limited, resource-limited clinical environments. This highlights why more annotation-free and computationally lightweight approaches are needed, which will encourage the development of further research on self-supervised learning.

The recent developments of Vision Transformer technologies (ViT) and self-supervised learning have accelerated medical imaging research. Wang et al. used SimCLR applied to lung CT by matching augmented patch views, and refined a DenseNet121 backbone using resultant embeddings. Even though this method enhanced the learning of representations, it depends on large-scale augmentations that are associated with the occurrence of anatomically unrealistic artifacts and longer training durations [22]. Zhou et al. used Momentum Contrast (MoCo) and ResNet50 encoder; they included a dynamic memory bank to supervise itself. Although it is effective on unlabeled datasets, MoCo depends on the large feature queues, which complicates its implementation in real-time clinical practice [23]. Li and Yang came up with a BYOL-based method that eliminates negative pairs using dual-network self-distillation. The model is, however, successful because of the correct momentum and predictor scheduling, which may result in instability unless well tuned [24]. Oquab et al. proposed DINOv2, which consists of self-distillation and masked pretraining on ViTs and showed good zero-shot performance in detectors [25]. Although these improvements have been made, the computational costs of transformer-based SSL methods are too large, hyperparameters are too sensitive, and the methods are not adequately tuned to lesion-level detection and classification tasks. Such restrictions bring to the fore the necessity of annotation-free SSL systems that enable precise nodule-level analysis yet can be handled by clinicians.

In general, the LUNA16 dataset has become the standard in self-supervised methods learning in detecting pulmonary nodules. Other papers that have employed LUNA16 have primarily concentrated on representation learning, without paying enough attention to nodule-level localization or interpretability of the learned features. These works, together, provide evidence that, despite their ability to generate strong representations, volumetric CNNs and transformer-based SSL models still require annotations, are expensive to compute, or lack analysis at the lesion level. This loophole emphasizes the necessity of the annotation-free SSL framework that compromises between the strong representation learning and the computational efficiency, and the nodule-level detection.

3. METHODOLOGY

This study presents an innovative self-supervised learning framework for identifying and classifying lung nodules from chest CT scans in the LUNA16 dataset. In Figure 1, we first extract deep feature maps from each LUNA16 CT slice using a SimCLR model pretrained via contrastive learning. It maximizes agreement between the augmented views of the same slice while separating others [26]. Next, we compute pairwise similarities among these representations to identify similar (positive) and dissimilar (negative) feature pairs, which are the basis for generating pseudo-labels without manual annotations. Pseudo-labels are then compared to the existing LUNA16 ground-truth annotations to remove low-confidence areas and narrow candidate proposals. Among these filtered candidates, we construct dynamic templates and use normalized cross-correlation (NCC) matching to retrieve a slice-wise similarity score map, which is computed efficiently [27] using fast integral-image algorithms. We came up with a new segmentation technique, which integrates normalized cross-correlation with k-means clustering to improve the segmentation process. Normally, the cluster is provided as input in the k-means clustering method. Although this method may work well in single images, which is subjective, it is not feasible when dealing with many images in a dataset and defining clusters (k) manually. Fixing this Figure in all images can have a drastic effect on the findings. In order to solve this, we use a dynamic clustering algorithm, which is a combination of normalized cross-correlation and k-means clustering. A Template matching method is used, in which normalized cross-correlation is done with the local mask to the full image to find matching signals as peaks.

These peaks indicate the number of clusters that are identified in each image, and the number is then forwarded to the k-means algorithm to further divide. The result of this segmentation is utilized to take out features with an extractor deprived of a Haralick feature extractor based on the GLCM, and the ensuing features are submitted to existing classifiers. This end-to-end pipeline applies LUNA16 dataset, which provides fast and automated analysis, which does not require manual labels to train the system. Validation Experimentally, our annotation-free framework performs with a high detection rate of 95.43 percent and at the same time the false positives are minimized. The ground-truth annotations were limited to template matching and performance benchmarking only to guarantee that the training process did not require manual supervision.

3.1 Dataset Curation and Preprocessing

The data set was the LUNA16 Lung nodules Analysis 2016 dataset that is publicly accessible [28]. It consists of 888 CT scans of 888 records of patients and is stored in MetaImage (.mhd) format and divided into 10 subsets. Figure 2 represents the sample images of the LUNA16 data set, which is also expected to test the lung nodule detection systems. Also, in Figure 1, the additional sample images of the LUNA16 dataset are presented. We obtain 1,186 annotated nodules, which are not involved in the training to ensure that they are considered at the point of final performance

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evaluation, since they are not part of the training to match the self-supervised foundation of our approach. The first stage in the pre-processing process is to transform the CT volumes into 2D axial slices. Once an image has been 2D converted, contrast enhancement and normalization are done to normalize the image intensity values when using different scans. These measures are necessary to ensure consistency in Hounsfield Unit (HU) mapping between sets of patients. The slices that are already pre-processed are then fed to the self-supervised pipeline, where unsupervised strategies are employed in learning the feature representations. These properties are subsequently used in downstream processes, such as dynamic k-means clustering, segmentation refinement, and candidate classification. The dataset was preprocessed, and at the slice level, it was partitioned randomly, 70 percent of the slices as training, 20 percent as validation, and 10 percent as testing.

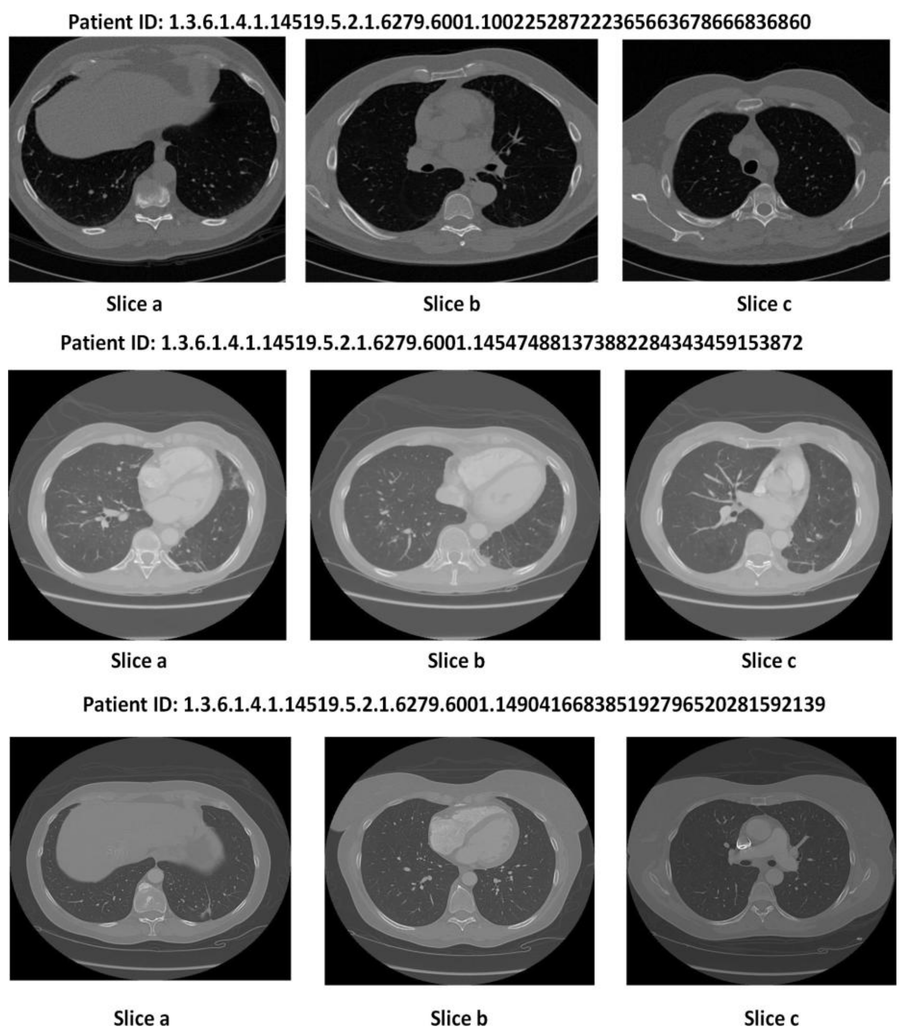

Figure 1. Sample 2D CT slices of the LUNA16 dataset, depicting different patient IDs and the axial views (e.g., LUNA16 a, b, c) of cut-through slices, used in the analysis of lung nodules.

3.2 Network Architecture

In order to elaborate further, the following sections will give a detailed, step-by-step dissection of the proposed framework. The stages of an entire pipeline, presented in Figure 2, i.e., feature extraction and pseudo-labeling, detection, and classification, are separated to demonstrate the underlying architecture and processing chain.

3.2.1 SSL Pretraining using SimCLR

In this phase, we utilize the SimCLR framework [26] with a ResNet-50 backbone [29] and a two-layer multilayer perceptron (MLP) projection head to derive robust feature embeddings from unlabeled 2D axial CT slices of the LUNA16 dataset. Unlike the original SimCLR formulation, which heavily relies on data augmentation to generate contrastive pairs, our approach capitalizes on the inherent slice-level variations within dataset to create positive and negative pairs. In the contrastive learning stage, positive and negative pairs were generated without the use of manual annotations. Positive pairs were defined as regions of interest (ROIs) or slices within the same CT scan that shared high structural similarity based on normalized cross-correlation (NCC) and feature-based template matching. Negative pairs were selected from spatially distant slices within the same scan or from different patients, ensuring that they did not represent the same nodule. We chose patches from anatomically similar and dissimilar regions, respectively. The model learn strong and distinctive features directly from the data without needing explicit labels. Training was carried out using the NT-Xent (Normalized Temperature-scale Cross Entropy) loss function, with key Hyperparameters including: batch size (128), learning rate (0.001 optimized using the adam algorithm), temperature parameter ($\tau = 0.5$), and spanning 200 epochs. Through the pretraining, we remove the projection head and store the encoder to generate high-dimensional feature maps to be used later. We will use a sample segment to capture the deep features of this segment to make it localizable. Then, the cosine similarities between the embeddings are computed in pairs and k-means clustering, such as DeepCluster [30], is used to produce pseudo-labels, which point to the potential nodule locations. To eliminate low-confidence clusters these pseudo-labels were cross-correlated with a small subset of LUNA16 annotations, which are only used in template matching and filtering. Notably, these annotations were never used in the training or pseudo-label creation stage on the entire dataset, which would essentially decrease false positives by eliminating clusters that do not meaningfully overlap the annotated nodule regions or get low similarity confidence scores. The process of filtering removes the patterns that are not nodular and the noise that is not desired. This process therefore provides one high confidence dynamic template.

3.2.2 Template matching

During the template-matching stage, we chose a nodule region on a representative CT slice (100 x 100 pixels) with a high confidence to use as our 'template' on which to match the remaining slices. This is a 100x100 pixel template, which is used to represent an area of about 70 mm x 70 mm in physical space, depending on the average spacing between voxels in LUNA 16 (0.7 mm/ pixel). With the nodules' size extending from 3 mm to 30 mm, the dimension fully represents the small and the large nodules along with their immediate tissue environment. The selected size is optimal in completeness, accuracy, and efficient computation in the template matching process. Pseudo-labels were used to identify the template regions, with ground-truth annotations only being used in a restricted ability to determine template quality. This makes training and detection annotation free.

All the target slices are converted to a normalized grayscale image in order to achieve uniform intensity scaling. After that, we are moving the template over the slice and computing the normalized cross-correlation (NCC) in each position, creating a similarity map. Rather than predefining the number of clusters (k), we actively determine it by determining the various peaks in the NCC map, each of which implies a high level of spatial fit between the input CT slice and the template.

In order to improve reliability, we find local maxima in the map of NCC responses that are above some threshold (e.g. 90 percent of the maximum world) and use non-maximum suppression to remove overlapping or spurious responses. The value of k dynamically for that slice is dependent on the number of these high-confidence peaks. Then we use k-means clustering on the spatial coordinates of these validated peaks to clustering them into coherent regions of interest (ROIs) without having to manually tweak the parameters used. Lastly, we refine the ROI clusters by another, localized NCC sweep to clean up the edges and the resulting segmented patches are distributed onto feature extraction and classification. This is a nonparametric, NCC-based k-means algorithm that removes the subjectivity of determining the number of clusters to use and fits well to the range of nodule appearances in the whole LUNA 16 dataset. NCC between the template u(x, y) and CT slice v(x, y) is calculated at every displacement (p, q). The NCC is a measure of the similarity of the template and the various regions of the bigger picture, as represented in Equation (1).

$$c_{vu}(p,q) = \sum_{x}\sum_{y}\big(v\,(x,y)u\,(x\,-\,p,y\,-\,q)\big) \qquad \dots\dots\dots(1)$$

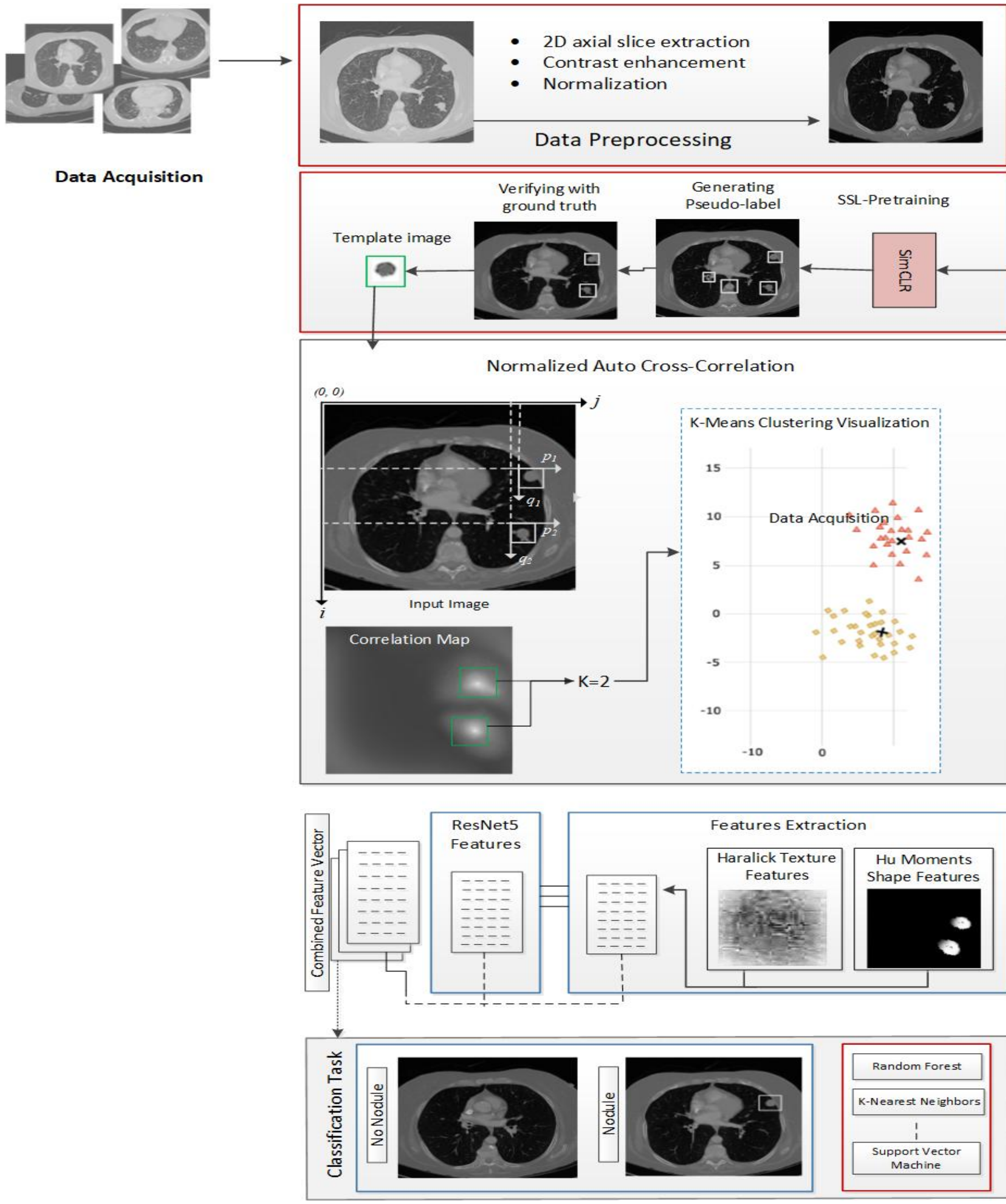

Figure 2. Self-supervised pipeline for lung nodule detection and classification without manual annotation

3.2.3 Improved segmentation model for dynamic k-Means clustering

Medical images are important in the analysis process, which involves detection and segmentation of the lesions or nodules. In CT scans of the lungs, proper segmentation is crucial in bringing about areas of interest like nodules, which might be present in the scans, by isolating them against the rest of the body. The adaptive, NCC-directed k-means clustering technique will be used to provide segmentation in our approach, since it will not demand a user-specified number of clusters. The traditional k-means separates pixels into k clusters depending on their closeness to predetermined centres. Nevertheless, correcting k on a large dataset of CT images may lead to under-segmentation or over-segmentation. In order to resolve this problem, we embark on template matching with normalized cross-correlation (NCC): a local nodule template is shifted through each slice to produce a similarity response map. High peaks in this map denote strong template matches, by measuring these highs, we are in a position to dynamically compute the appropriate k of that image. This automatically identified k is then used in k-means algorithm which clusters the spatial locations of the peaks of the NCC into coherent regions of interest (ROI). This NCC-based dynamic clustering is flexible to changes in nodule appearance and quantity across slices, which makes the segmentation process easier as the subjective parameter decisions are not required. Finally, the segmented ROIs are sent to feature extraction which incorporates both handcrafted descriptors and deep embeddings during the classification stage.

3.2.4 Feature extraction

After the segmentation stage is done, we derive texture and shape features of the clustering regions to further categorize lung nodules. Extracting the texture features entails the use of Gray-Level Co-occurrence Matrices (GLCM) [31], which is extensively utilized in medical imaging because it is effective in defining the spatial relationship between pixel intensity pairs within the specific offsets [32]. Out of every GLCM, we extract significant descriptors including energy, entropy, correlation, and homogeneity [33], that indicate consistency, randomness, and structural patterns of nodule textures [34]. We also derive additional measures, including contrast and variance, to reflect the local intensity changes in the images, thus facilitating the identification of subtle radiographic details in the CT images [35]. To find shape features, we apply to the central image moments the seven moment invariants of Hu, which are invariant under translation, rotation, and scaling [36]. The invariants have proved useful in geometry of the segmented regions, as they help to distinguish between circular nodules and long vessels by compacting the object contours into a small feature vector [37]. The resources of OpenCV show how efficient they are in terms of computation, thus they are appropriate in processing large volumes of 2D CT [38]. GLCM and Hu moment combination provides both a textural analysis and structural analysis of lung nodules. The internal pixel variation is modeled by texture features, whereas the shape and boundary geometry is modeled by Hu moments. This combination provides a better classification as both the appearance and morphology are considered to provide a nodule differentiation.

3.2.5 Classification

We evaluated six standard supervised classifiers in the framework of the SSL in order to label nodules and non-nodules using our joint deep (SimCLR) and hand-crafted (GLCM, Haralick, and Hu moments) features.

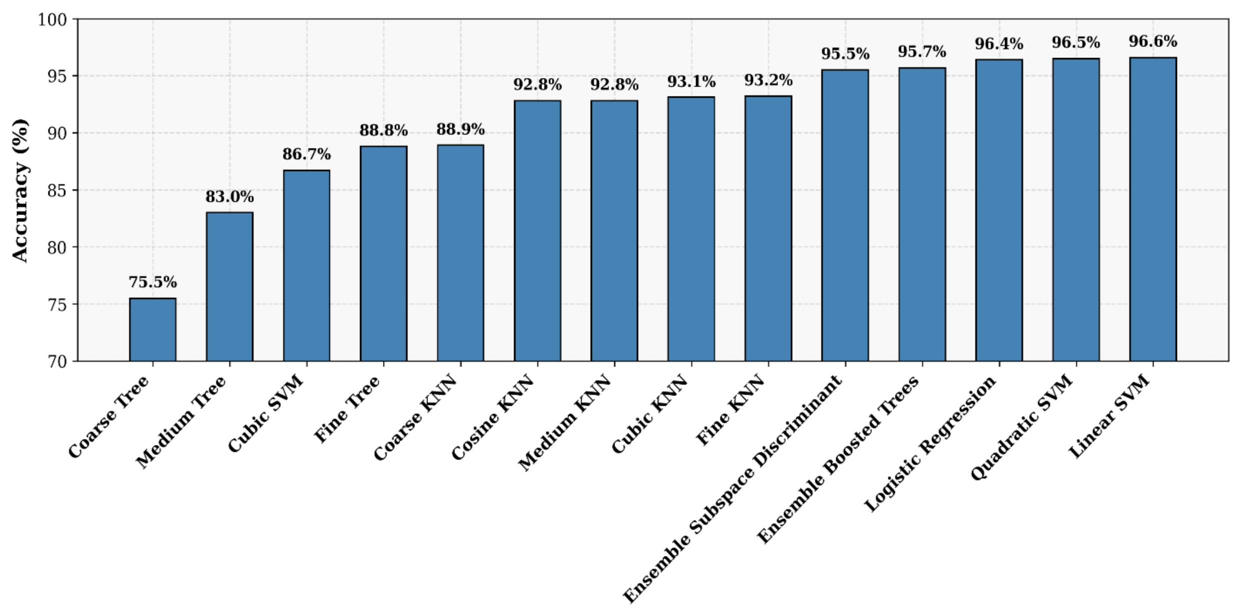

Figure 3. Representing the various classifiers' performance

Linear, cubic, and quadratic kernels that are Support Vector Machines (SVM) [39] were tested to ascertain the impact on high-dimensional radionics features of the selection of the kernel. K-Nearest Neighbors (KNN) was employed, whereby Euclidean distance (fine, medium, coarse) and cosine distance were used to determine the most suitable definitions of neighborhoods in feature space [40]. Decision Trees (DT) were used in shallow (coarse) configurations, medium, and deep (fine) configurations to build rule-based classification boundaries that can be interpreted [41]. We also used Ensemble Models such as Random Forests (bagged trees) [42] to achieve the strength of aggregating weak learners. The Logistic Regression served as a baseline linear regression classifier that probabilistically output results, which served to measure decision thresholds [43]. Each of these models was trained and evaluated on the same set of features by a stratified cross-validation; the accuracy, sensitivity, and specificity metrics in each case are given in Table 2, with SVM-linear achieving the best overall accuracy, and KNN-cosine performing better in detecting small nodules [44]. Figure 3 represents the performance of the classifier as a whole in the classification of nodules and non-nodules.

3.2.6 Evaluation parameters

We used the usual performance metrics to estimate the performance of the proposed classification parameters as given below:

Accuracy: Accuracy is also referred to as Rand index and is employed to visualize the number of correct predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{2}$$

Precision: The number of true positive results divided by the total positive results is known as precision score.

$$precision = \frac{\mathrm{TP}}{TP + FP} \tag{3}$$

Specificity/Recall: Recall or sensitivity is the proportion of the true positive results to the total of the true positives and the false negatives.

$$Recall = \frac{\mathrm{TP}}{TP + FN} \tag{4}$$

F1 Score: It is the weighted average of the precision and the recall scores of a model. The metric of accuracy is the ratio of correct predictions of a model on the whole data set.

$$F1\ score = 2 \times \frac{precision \times Recall}{precision + Recall} \tag{5}$$

4. EXPERIMENTAL RESULTS AND DISCUSSION

This section provides a summary of the results of our SSL method. The experimental method has two stages. We first use NCC to analyze the unlabeled CT scans of the LUNA16 dataset to find possible nodule regions. We then use Adaptive K-means clustering to do this step automatically, without any manual work to identify these regions. The second stage involves feature extraction on each of the segmented regions and features are produced by combining handcrafted descriptors (including Haralick texture features and Hu moments) with those generated at a deep level with the ResNet-50 model. This all-inclusive set of properties educates different classifiers such that it can precisely classify nodules without the need to utilize labeled information.

4.1 Classification Results

The ResNet50 architecture has an inbuilt classification layer, which makes us use the output of the architecture as a feature extractor without an external classifier. Conversely, when evaluating the performance of a concatenated feature set of deep feature ResNet50 plus handcrafted features derived by using normalized cross-correlation-based k-means segmentation, we tested their performance on a few machine learning classifiers to understand their ability to differentiate nodules and non-nodules. Table 1 provides the overall comparison of the performance of the different classes of classifiers, the Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Logistic Regression, and Ensemble models. The greatest Accuracies in the linear and quadratic SVMS were the highest at 96.6 and 96.5, respectively. These findings depict how margin-based classifiers are efficient in dealing with high-dimensional fused feature spaces. In the same manner, the Logistic Regression also did an impressive job with an accuracy of 96.4, which indicates that it is efficient when dealing with linearly separable representations produced by integrated features.

Ensemble classifiers, e.g., Boosted Tree and Subspace Discriminants, turned out to be the best ones, with scores of 95.5 and 95.7, respectively. This points out the positive effect of model diversity and averaging in minimizing variance and enhancing generalization. Fine and Cubic KNN variants of the KNN family achieved the highest accuracy of 93.2% and 93.1% with other variants achieving a maximum accuracy of approximately 92.8%. By contrast, tree-based classifiers, such as Coarse, Medium, and Fine Trees, which demonstrated a little lower performance, between 75 and 88, may be explained by high sensitivity to noise as well as failure to capture interactions between features. In general, margin maximization (SVM) and ensemble-based classifiers were the best performers in nodule classification when using a rich combination of learned features, ResNet50, NCC template matching, and clustering, which are trained in a self-supervised and annotation-free fashion. The quantitative summaries presented in Table 2 and Figure 3 support the conception according to which our proposed pipeline is effective in classifying lung nodules precisely.

Table 1. Performance metrics of classification models

Classifier	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-score (%)**	**AUC (%)**
Co Coarse KNN	88.9	93.8	83.4	88.3	0.96
Cosine KNN	92.8	99.2	87.8	93.2	0.99
Cubic KNN	93.1	94.0	93.1	93.6	0.99
Fine KNN	93.2	95.0	93.2	94.1	0.99
Medium KNN	92.8	94.0	93.3	93.6	0.99
Coarse Tree	75.5	84.0	75.6	79.5	0.79
Medium Tree	83.0	84.0	83.1	83.5	0.88
Fine Tree	88.8	90.0	88.8	89.4	0.92
Quadratic SVM	96.5	96.0	96.6	96.3	0.99
Linear SVM	96.6	96.0	96.6	96.3	0.99
Logistic Regression	96.4	96.0	96.4	96.2	0.99
Ensemble Boosted Tree	95.7	95.0	95.7	95.3	1.00
Ensemble Subspace Discriminant	95.5	96.0	95.6	95.8	0.99

4.1.1 Linear SVM

The linear Support Vector machine (SVM) finds a plane maximizing the distance between classes. By applying SVM, a high accuracy of 96.6% and an AUC of 0.99 were obtained. The ROC curve exhibits the high discriminatory potential of the model whereas the confusion table indicates that there are very few misclassifications, which highlights its usefulness in the classification of lung nodules as it is illustrated in Figures 4 and 5.

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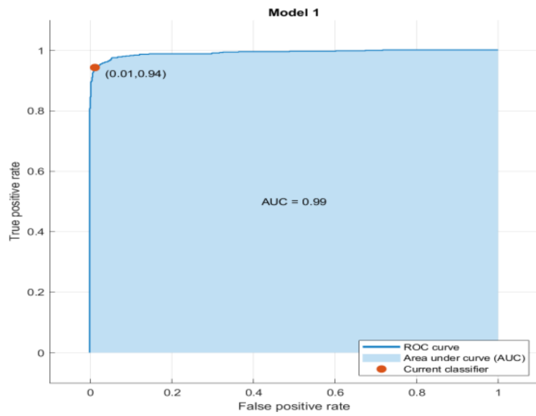

Figure 4. ROC Curve representing the performance of the Linear Support Vector Machine

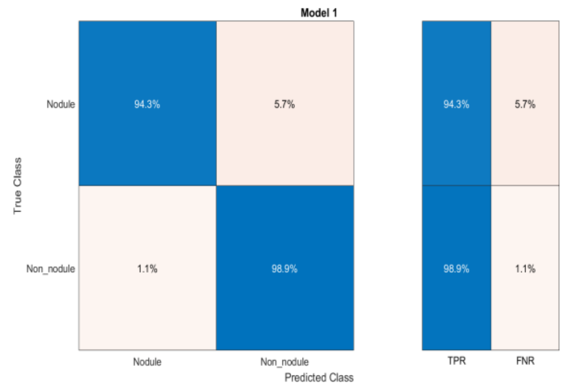

Figure 5. The Confusion Matrix shows the Performance of Linear Support Vector Machine

4.1.2 Logistic Regression

Logistic Regression estimates the probability of belonging to a class using a logistic function. It achieved an accuracy of 96.4% and an AUC of 0.99 in our experiments. The ROC curve demonstrates its robustness across various thresholds, and the confusion matrix shows a balanced performance with low false positives and negatives in Figures 6 and 7.

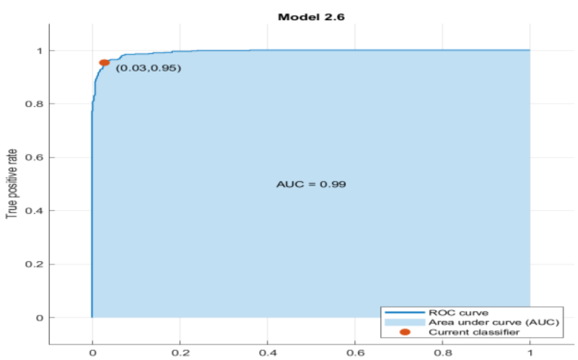

Figure 6. ROC Curve shows the performance of Logistic Regression

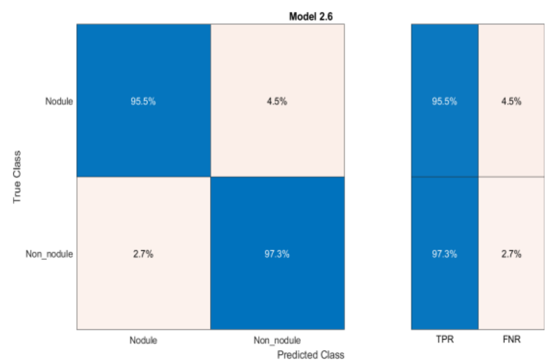

Figure 7. Confusion Matrix Display for Logistic Regression Performance

4.1.3 Ensemble Boosted Tree

Ensemble Boosted Tree combines multiple weak learners to form a strong classifier. The accuracy of 95.7% and an AUC of 1.00 were achieved. ROC curve confirms its outstanding performance in Figures 8 and 9, and the confusion matrix reflects its effectiveness, though with slightly more misclassifications than the top two classifiers.

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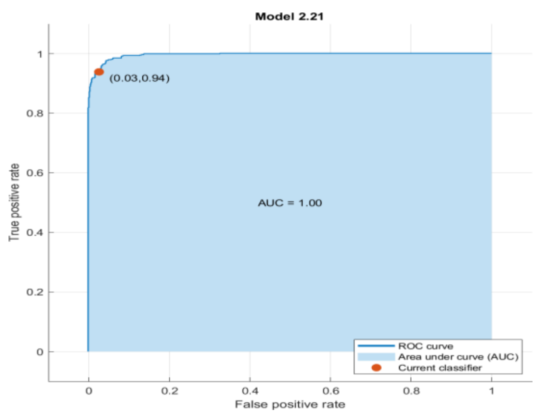

Figure 8. ROC Curve Display for Ensemble Boosted Tree Performance

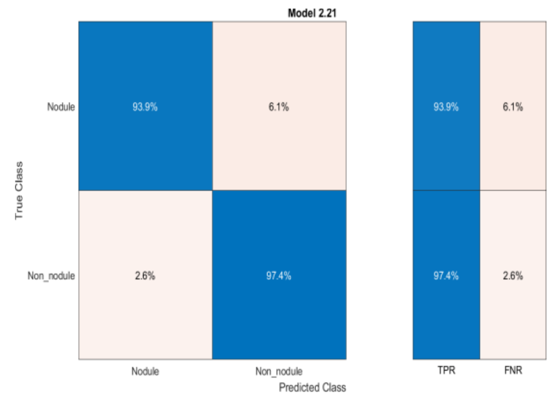

Figure 9. Confusion Matrix display for ensemble Boosted Tree performance

4.2 Comparative Analysis

In Table 2, although MAEMC NET [45] and Fine-Grained SSL [46] achieve over 93% accuracy, the former is tailored to specific nodule types. At the same time, the latter requires extensive pre-training on thousands of nodules, which can limit their applicability to broader clinical datasets, 1200 and 5478 CT scans, respectively. STLF VA's attention-driven transfer learning [47] does not provide the AUC value in their research. Similarly, SSTL DA's domain-adaptation SSL [48] enhances feature discrimination across various CT volumes of Luna16. However, they incur high computational overhead and do not inherently provide unsupervised region segmentation. ERB- Net's ensemble of 3D CNNs [49] offers robustness against nodule heterogeneity and does not report the AUC value in their study, yet it significantly increases model complexity and inference time. To address these limitations, we explored an annotation-free pipeline that utilizes normalized cross-correlation template matching and dynamically adaptive K-means clustering to generate pseudo-labels and segment candidate regions, then fuse handcrafted Haralick and Hu moment descriptors and ResNet 50 features on the Luna16 challenging dataset. Our method achieves 95.43% accuracy and 96.5% AUC with only ResNet50, as shown in Figure 10. Additionally, it achieves better results by fusing ResNet50 and NCC, as well as K-means Clustering methods, demonstrating an efficient and scalable method for robust lung nodule localization and classification.

Table 2. Comparison with Baseline Methods

Method	SSL Strategy	Samples (CT)	Acc.(%)	AUC(%)
[45] MAEMC-NET	Hybrid Contrastive+Masked AE	1,200	93.7	96.2
[46] Fine-Grained SSL	Fine-grained pre-training	5,478	93.4	96.4
[47] STLF-VA	Attention-driven TL	1,018	95.8	—
[48] SSTL-DA	Domain-adaptation SSL	1,018	91.1	95.8
[49] ERBNet	Ensemble 3D CNN	1,500	95.0	—
Proposed	ResNet50	941	95.43	96.5
	ResNet50+NCC+ Kmeans	941	96.06	0.99

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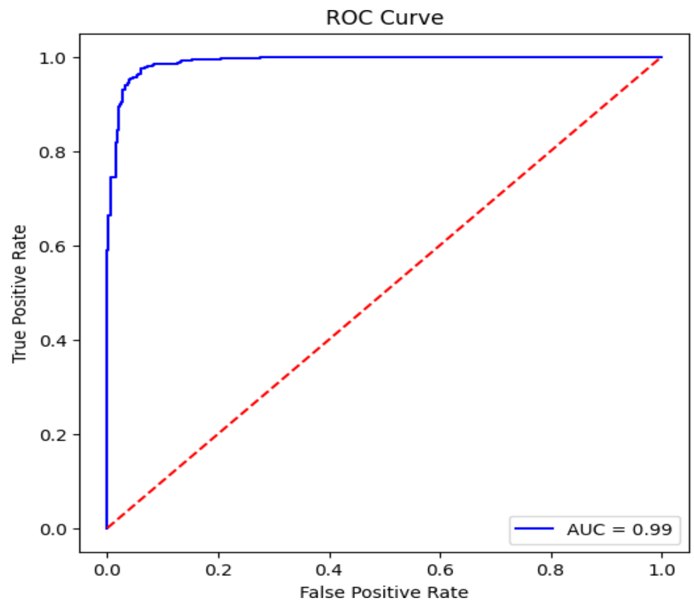

Figure 10. AUC curve representing the ResNet50 performance

4.3 Ablations study

In this section, we evaluate the role of each significant constituent of our pipeline by an ablation study that is exemplified in Table 3. First, we consider deep features and only train ResNet embeddings of size 50, which provides a high baseline, as well as shows the possibility of improving results. Next, we separately assess the impact of handcrafted descriptors, including GLCM-based Haralick features and Hu moments and show that although they are useful in terms of texture and shape information, they are not sufficient. To study the segmentation process with NCC-based template matching and adaptive K-means clustering as the tool to form basic region-based features, we observe moderate advances in comparison to handcrafted features. Finally, our full pipeline, with all three elements (deep, handmade, and segmentation-derived feature), has the best accuracy, and AUC, showing the critical and interactive roles of each of the modules in successful, annotation-free lung nodule detection and classification with lower false positives.

Table 3. Ablation Study on LUNA16 based on Component Contributions

Model Variant	**Accuracy (%)**	**AUC (%)**
ResNet-50 features only	92.1	94.5
Handcrafted (Haralick + Hu) only	89.8	92.0
NCC + K-means segmentation only	90.5	93.2
Full pipeline (Fusion)	95.43	96.5

5. CONCLUSION

We have come up with self-supervised frame work of the detection and classification of lung nodules in CT scans without the use of annotations. The suggested pipeline is a fusion of normalized cross-correlation template matching and dynamically adaptive K-means clustering of candidate nodules automatically without using any manual segmentation masks. Deep embeddings are used based on a ResNet-50 backbone to create pseudo-labels during pre-training and help classifiers during fine-tuning along with handcrafted descriptors (Hu moments and GLCM-based Haralick features). Experimental findings on our LUNA16 dataset reveal that our framework is of high accuracy, sensitivity as well as specificity but it has a great reduction of reliance on labeled data. Notably, template validation and benchmarking were performed only with a few annotations in order to make the framework annotation-free. Furthermore, the objective of our pipeline one of the primary ones was to reduce false positives, which is a typical issue with CAD systems, which is efficiently dealt with by our clustering and classification system based on similarity.

Although the nodules can be reliably detected and classified using our framework, it lacks full-pixel-level segmentation of nodules. Future studies will focus on elaborating the framework to accurate volumetric segmentation, perform cross-validation on a per-patient basis, and apply the technique on high-volume multi-centric data in order to enhance better generalizability and reliability. Another area of expansion that we intend to do is to expand its use to other clinical applications other than lung CT scans by looking into multi-modal imaging like the use of PET-CT fusion and other clinical applications. Finally, the algorithmic enhancements will be done to make it scalable and allow using it in real-time clinics. Such improvements will be useful in the creation of early lung cancer detection and diagnosis tools that are practical without the use of annotations.

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